

Project Design Phase-I Proposed Solution Template

Date	12/03/2026
Team ID	SWUID20264931
Project Name	Electric vehicle Charge and Range
Maximum Marks	

Proposed Solution Template:

Project team shall fill the following information in proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Addressing the limited range of electric vehicles and the lengthy charging times as a barrier to widespread EV adoption.
2.	Idea / Solution description	Develop a fast-charging infrastructure and battery management system to significantly reduce EV charging time and extend the operational range of electric vehicles.
3.	Novelty / Uniqueness	Implement ultra-fast charging stations capable of charging EVs to 80% in under 15 minutes and a smart battery management system for optimal range enhancement.
4.	Social Impact / Customer Satisfaction	By reducing charging time and extending range, the solution makes EVs more practical and convenient for daily use, accelerating the transition to sustainable transportation.
5.	Business Model (Revenue Model)	Revenue generated through pay-per-use fast-charging stations, subscription models for unlimited charging, and partnerships with electric fleet operators.
6.	Scalability of the Solution	Scalable network of fast-charging stations that can be expanded to urban, suburban, and highway locations. Adaptable technology that can be integrated with renewable energy sources.